Hybrid Model for Early Detection of Diabetic Retinopathy using Deep Learning and Machine Learning

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Abstract. Diabetic retinopathy is a serious complication of diabetes mellitus that can lead to vision loss if not diagnosed early. The aim of this study was to identify diabetic retinopathy using a robust model. The methodology employed was based on six phases: dataset acquisition, preprocessing (CLAHE, standardization and grayscale), feature extraction (SIFT, ORB, HOG), machine learning models (RF, KNN, SVM), deep learning models (BOTNet, CoAtNet, ViT, Swin Transformer, MobileViT and PiT), a hybrid approach and model evaluation (accuracy, precision, sensitivity and F1-Score). The best results demonstrate that the PiT model optimized with the hyperparameters (batch_size: 32, epochs: 15, learning_rate: 0.001 and optimize: SGD) obtained the best performance with an accuracy of 98.70%, precision of 99.11%, recall of 98.23%, and F1-score of 98.27%. In conclusion, the robust deep-learning-based model with hyperparameter adjustment significantly improved the classification of diabetic retinopathy.

Keywords: Diabetic Retinopathy, Deep Learning, Machine Learning, Hybrid Model, Preprocessing Techniques, Feature extraction, PiT, Hyperparameters.

1 Introduction

Diabetic retinopathy (DR) is a complication of diabetes mellitus (DM) that causes progressive damage to the blood vessels of the retina, which, if not detected and treated in a timely manner, can lead to irreversible vision loss [1], [2]. This disease manifests through microaneurysms, intraretinal hemorrhages, and hard exudates [3]. Elevated blood glucose levels induce thickening of the basement membrane, loss of pericytes, and degeneration of endothelial cells in retinal capillaries, leading to irregular blood flow and disruption of the blood-retinal barrier [4]. DR is one of the leading causes of preventable blindness and vision loss in adults worldwide, affecting both developed and developing countries, with significant variations in prevalence among regions [5]. This situation is especially alarming in areas with rapid population growth and demographic changes, where the increase in DM has significantly increased the burden of related eye diseases [6].

According to the World Health Organization (WHO), DR is among the leading causes of blindness and visual impairment worldwide. It is estimated that more than 2.2 billion people suffer from some type of visual impairment, and at least 1 billion of these cases have not received adequate care, even though they could have been prevented or

treated in a timely manner [7]. In a study conducted in 2020, the prevalence of DM in adults in Mexico was 13.7%, which significantly increases the risk of developing complications such as DR and other associated diseases [8]. In Peru, the prevalence of DM in the adult population was estimated at 7%, with less than 30% of cases under adequate disease control. This problem is more prevalent in urban areas, such as Lima (8.2%), compared to the Andean (4.5%) and Amazonian (3.5%) regions. In addition, it is estimated that DR affects 23%–30% of patients with type 2 diabetes, doubling the risk of visual impairment in those without this complication [9]. According to the National Institute of Statistics and Informatics (INEI), in 2020, 4.5% of the Peruvian population aged 15 years or older had been diagnosed with DM, reflecting the considerable burden that this disease represents in the country [10].

Healthcare professionals use various techniques to diagnose diabetic retinopathy (DR) in a timely manner. Some of these methods are noninvasive, such as the dilated eye exam and optical coherence tomography (OCT), which uses light instead of sound to capture detailed images of retinal structures [11]. However, in advanced cases or when a more detailed analysis of retinal blood vessels is required, fluorescein angiography, an invasive procedure involving the injection of a dye into the bloodstream—may be necessary. Although effective, this technique can have side effects, including nausea and allergic reactions [12]. Despite their utility, many of these diagnostic tools are expensive and may not be readily accessible in resource-limited settings, posing significant challenges for the early detection and management of DR.

In recent years, convolutional neural networks (CNNs) in the field of Deep Learning (DL) and Machine Learning (ML) classification algorithms have been shown to be significantly more effective than traditional approaches, enabling more accurate and efficient classification of DR. In a previous study [13], a framework based on ML and DL techniques using the DCSGAN (Diabetes Classification Model Based on Generative Adversarial Networks) model is proposed, which demonstrated high accuracy in the classification of patients with diabetes. In turn, in [14], a detailed analysis highlighted how advances in ML and DL have transformed diabetes detection, employing algorithms such as CNNs, support vector machines (SVM), decision trees (DT), and random forest (RF) to classify patients and predict the probability of developing this disease.

Despite recent advances, there remains a need to develop innovative models to support medical diagnosis, especially in resource-limited regions. This study proposes a robust approach using DL, ML, and a hybrid approach to optimize early detection of diabetic retinopathy.

2 Related Work

Das et al. [15] described DR as an ocular complication associated with DM, characterized by inflammation and rupture of retinal blood vessels. These alterations cause lesions such as microaneurysms, hemorrhages, and exudates, which can progress to progressive and, in severe cases, total vision loss. The authors emphasize that early diagnosis is essential to prevent severe complications, such as blindness, but acknowledge that traditional manual methods for DR detection have significant limitations. To

overcome these limitations, this study proposes an intelligent DL-based system that automatically analyzes fundus images using CNNs. The proposed method facilitates the hierarchical extraction of complex features and accurate classification of different stages of DR. The proposed methodology comprises 4 phases: Image dataset acquisition (Kaggle EyePACS), Image preprocessing, model training (VGG-16, VGG-19, Xception, InceptionV3, InceptionResNetV2, MobileNet, Mobile-NetV2, EfficientNet B0-B7, DenseNet121, DenseNet169, DenseNet201, ResNet50, ResNet50V2, ResNet101, ResNet101V2, ResNet152, ResNet152V2, NASNetLarge and NASNetMobile), and model evaluation. The results of this study revealed that DenseNet201 achieved the highest accuracy during training (99.58%), while EfficientNetB4 excelled in validation and testing with an accuracy of 79.11%, thereby positioning itself as the most efficient model for practical applications. The findings underscore the ability of automated DL-based systems to optimize early detection of DR, improving diagnostic accuracy and significantly reducing the risk of blindness, especially in resource-limited settings.

Alghamdi et al. [16] highlighted that DR is a frequent complication of diabetes, affecting approximately one-third of patients and positioning itself as the leading cause of preventable vision loss in people of working age. The disease arises due to damage to the retinal blood vessels and, although initially asymptomatic, progresses to stages that seriously jeopardize vision. Traditionally, DR is detected by manual analysis of fundus photographs by trained specialists, who assess the presence and severity of characteristic lesions such as microaneurysms, hemorrhages, cotton-wool spots, and exudates. However, the increasing global prevalence of diabetes and limited availability of specialists render this approach impractical. In this context, artificial intelligence (AI) algorithms, particularly those based on DL, have shown great potential in medical imaging, facilitating the detection, diagnosis, and prognosis of various pathologies. The main objective of this study was to evaluate advanced DL models for the automatic detection of DR using fundus images, highlighting the use of explainability techniques, such as Grad-CAM, to interpret model decisions. The adopted methodology included 4 phases: Dataset acquisition (Kaggle EyePACS); image preprocessing; model training (VGG-16, ResNet-18 and DenseNet-121), and model evaluation (Precision, Recall, F1-Score). The best results obtained were 87% Precision, 71% recall, and 73% F1-score. The results showed that the VGG-16 model performed best in terms of accuracy (87%), with an F1-Score of 73% and better generalization ability. Although DenseNet-121 achieved comparable values, it was overfitted during training. The authors concluded that although the models were able to identify patterns associated with DR, further research is needed to improve the interpretability of decisions, especially in the medical setting where transparency is essential for the clinical adoption of these technologies.

In Mohanty et al.'s study [17], DR was reported to be a metabolic disease characterized by elevated blood sugar levels that over time affected multiple organs, including the retina of the eye. The retina, which captures light and transmits signals to the brain, is critical for vision. Early and regular detection of DR using computer-aided diagnosis (CAD) systems is crucial, although this traditional process is often slow, error-prone, and, at times, uncomfortable for patients. DL architecture, especially CNNs, have emerged as effective image classification solutions. The purpose of this study was to

evaluate two approaches: a hybrid model combining VGG16 with the XGBoost classifier and the DenseNet121 model, to detect and classify DR, and to compare them with other methods to identify the most efficient architecture. The proposed methodology was structured in four phases: dataset acquisition (APTOS 2019 Blindness Detection); image preprocessing (balancing and resolution adjustment techniques); model training (VGG16, DenseNet121 and XGBoost), and model evaluation (Accuracy). The results demonstrate that DenseNet121 achieved an accuracy of 97.30%, which indicates that it is the most effective model. The hybrid model achieved an accuracy of 79.50%, which is higher than that of previous approaches. This study concluded that DL architectures, in particular, DenseNet121, are highly effective and can play a crucial role in DR detection and classification, improving diagnostic efficiency and accuracy, especially in clinical settings.

In the research conducted by Ayala et al. [18] DR was described as an ocular disease that affects the retinal blood vessels, damaging them due to metabolic disturbances related to elevated blood glucose levels. This condition usually develops in patients with undiagnosed diabetes for at least 10 years. Although DR can be prevented through regular health examinations and appropriate treatment, early diagnosis remains challenging. Recently, advanced DL techniques have been shown to be promising tools for the detection and classification of DR in different stages, surpassing the capabilities of the human eye in certain classification tasks. The objective of this study was to employ the DenseNet121 architecture, optimized by transfer learning, to automatically detect and classify DR using two datasets. The proposed methodology was structured in four phases: acquisition of fundus images (APTOS and Messidor), preprocessing (central region cropping and resizing to 224x224 pixels), model training, and model evaluation (accuracy, Recall and F1-Score). The results demonstrate that the model achieved variable performance depending on the class. During validation with the APTOS ensemble, the accuracy was highest for the "No DR" class (0.96) and lowest for "Severe NPDR" (0.40) and "PDR" (0.46). The results indicated that DenseNet121 significantly improved the automatic detection of DR, providing faster and more reliable results than traditional methods. However, it emphasizes the need for a standard dataset to facilitate comparison between different proposals and improve model generalizability.

Finally, in the research of Gill et al. [19] described DR as a common complication of diabetes that, in most cases, is not detected until it reaches advanced stages. The various abnormalities caused by RD include red lesions, such as microaneurysms (MA), intraretinal hemorrhages, exudates (EX), and cotton-wool spots. The manual diagnosis of this condition is laborious and costly and requires highly specialized personnel with experience in the interpretation of medical images. In this context, automatic DL-based techniques, such as pretrained CNN models, are promising tools to improve diagnosis accuracy and speed. The aim of this study was to develop an automated system to detect and diagnose DR in fundus images using a hybrid approach combining GoogleNet and ResNet-18 for feature extraction, followed by classifiers such as SVM, RF, RBF, and NB. The proposed method improves accuracy and reduces the errors associated with medical diagnosis. The proposed methodology consists of four phases: database acquisition (APTOS and Messidor); image preprocessing (resizing to 224x224 pixels and normalization); CNN hybrid feature extraction (GoogleNet and ResNet18);

model training (RF, SVM, RBF, NB); model evaluation (Accuracy, Precision, Recall, F-Measure). The results demonstrate accuracy of 97.80% for binary classification and 89.29% for multiclass classification using the SVM classifier. These results indicate that advanced stages of DR are easier to classify, whereas mild and moderate stages present greater challenges due to interclass similarities. The authors concluded that the proposed system is effective for the early detection of DR and outperforms previous research in terms of diagnostic accuracy and efficiency.

3 Methodology

The proposed methodology, based on previous studies [20], [21], [22], [23], [24], [25], consists of six phases: dataset acquisition, preprocessing, feature extraction, deep learning model application, machine learning model application, hybrid approach application, and model evaluation. The methodology is illustrated in Fig. 1.

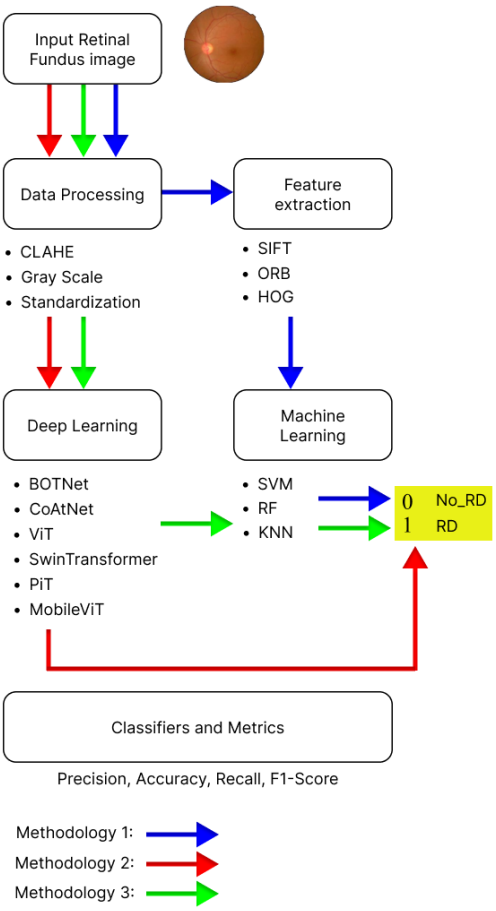


Fig. 1. Proposed methodology.

3.1 Dataset Acquisition

Parisa Karimi Darabi's "Diagnosis of Diabetic Retinopathy" database was selected for its comprehensive collection of high-resolution retinal images evaluated by a physician for the presence of diabetic retinopathy. The data set was obtained from https://www.kaggle.com/datasets/pkdarabi/diagnosis-of-diabetic-retinopathy/data and is organized into three main directories: train (73.15%), test (18.71%), and valid (8.14%). Each of these directories is subdivided into two folders: DR (images with diabetic retinopathy) and No_DR (images without diabetic retinopathy). The images have a resolution of 224 x 224 pixels in JPG format and are classified into two categories: Class 0 (No_DR) and Class 1 (DR).

3.2 Preprocessing

Table 1 shows the activities involved in the preprocessing phase.

Table 1. Dataset preprocessing activities

Activity	Description	Tasks
Apply CLAHE	Improves the contrast of images, especially in areas with low contrast, through adaptive histogram equalization.	Load the original (unprocessed) images.
		Apply the CLAHE algorithm to the images.
		Save the preprocessed images in the corresponding folder structure.
Image Standardization	Adjusts the image data by subtracting the mean and dividing by the standard deviation, ensuring comparable values.	Load the processed images.
		Apply the standardization technique.
Grayscale Conversion	Converts the images to grayscale to reduce dimensionality and highlight the relevant structural details of the retina.	Load the processed images.
		Apply the grayscale conversion.

Apply CLAHE: Contrast Limited Adaptive Histogram Equalization (CLAHE) is an advanced histogram equalization technique that enhances contrast in digital images while minimizing noise amplification. Unlike global histogram equalization, CLAHE divides the image into small regions called "tiles" and locally applies histogram equalization to each region. To avoid excessive contrast amplification, which can cause visual artifacts, CLAHE imposes a contrast limit using a parameter called "clip limit". This value restricts the maximum height of the peaks in the histogram before redistributing the intensities in the image. Finally, interpolation between regions is used to smooth transitions and ensure a consistent appearance in the processed image. Given these features, CLAHE is ideal for enhancing medical images, low-light photography, and scenarios in which details must be highlighted while maintaining visual quality [26].

Image Standardization: Standardization with mean and standard deviation is a technique commonly used in deep learning data preprocessing to adjust feature values (in this case, image pixels) to a more appropriate scale for classification algorithms. According to Goodfellow et al. [27], this standardization ensures that each input feature has a distribution with zero mean and unit standard deviation. This allows the neural networks to process images more efficiently, preventing features with large scales from dominating over others and promoting better convergence during training.

$$I_{estand} = \frac{I - \mu}{\sigma} \tag{1}$$

Where:

- $I_{estand:}$ The standardized value of the pixel or data.
- *I:* Is the original value of the pixel or data.
- μ: Is the mean of the values of all the pixels (or data) in the set.
- σ: Is the standard deviation of the values in the set.

Grayscale Conversion: This conversion is particularly useful in medical images, such as X-rays, CT scans, or retinal images, where color does not provide additional diagnostic information. In these applications, grayscale images highlight relevant structures and facilitate the extraction of important features for image classification and segmentation [28].

Fig. 2 shows the stages of preprocessing applied to fundus images. The original, unprocessed image first undergoes contrast adjustment using CLAHE. Subsequently, depending on the methodology used, the images are transformed to grayscale (Methodology 1) or standardized (Methodologies 2 and 3).

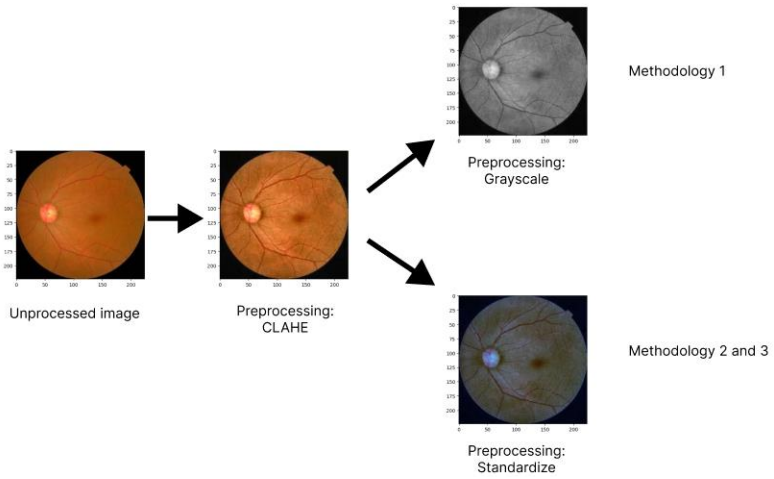


Fig. 2. Preprocessed images.

3.3 Feature extraction

In this phase, three feature extraction techniques were applied: SIFT, ORB and HOG. Table 2 shows the activities of the feature extraction phase.

Table 2. Feature extraction activities

Activity	Description	Tasks
Feature extraction with SIFT	Algorithms to detect key points in images and extract descriptors invariant to scale, rotation and illumination changes.	Load the preprocessed images.
		Apply SIFT
		Save the extracted features
Feature extraction with ORB	Fast and efficient algorithm that combines FAST keypoint detection with BRIEF descriptors, optimized for real-time applications.	Load the preprocessed images.
		Apply ORB
		Save the extracted features.
Feature extraction with HOG	Method representing the distribution of orientation gradients to capture patterns and shapes in images, which is useful for classification.	Load the preprocessed images.
		Apply HOG.
		Save the extracted features.

Feature extraction with SIFT: The SIFT algorithm is robust for detecting and describing distinctive key points in images, ensuring invariance to scale, rotation, and partial illumination changes. The process begins with the construction of a Gaussian scale representation to analyze the image at multiple scales. Differences in Gaussian distribution (DoG) were calculated to highlight stable key points, which were identified as local maxima or minima. To ensure rotational invariance, each key point is assigned an orientation based on the local gradient directions. Finally, a descriptor that summarizes the magnitudes and orientations of the gradients within a local region is generated, resulting in a 128-dimensional feature vector. These distinctive features allow SIFT to perform reliable image matching even in the presence of occlusion, noise, or clutter [29].

Feature extraction with ORB: This efficient technique detects and describes key points in images. The proposed method combines the FAST corner detector, which identifies key points by fast intensity comparisons between pixels, with the BRIEF descriptor, which generates binary representations of the local neighborhoods around each key point. To guarantee rotation invariance, ORB assigns an orientation to each key point based on the intensity moments in the neighborhood. In addition, it introduces enhancements, such as the use of oriented (rotated) BRIEF and a classification scheme to select keypoint with higher discrimination. These features make ORB fast, computationally efficient, and robust to rotations, scale changes, and noise, making it an ideal choice for real-time applications [30].

Feature extraction with HOG: This technique is widely used for feature extraction in computer vision, especially in object detection tasks such as pedestrian recognition. The proposed method captures information about the distribution of intensity gradients and their orientations in local regions of an image. First, the image was normalized to reduce the effects of illumination and contrast variations. The gradient of each pixel

was calculated in terms of magnitude and direction. The image is divided into small cells, and within each cell a histogram is constructed that groups the gradient orientations over a range of angles (e.g., 0° to 180° or 0° to 360°). To improve the invariance of the illumination changes, the histograms were normalized using overlapping blocks. The final descriptor is obtained by concatenating the histograms of all cells and generating a vector representing the image or region. This technique is notable for its robustness against small shape deformations and illumination changes, making it ideal for detecting distinctive visual patterns in images [31].

3.4 Application of Machine Learning Models

In this phase, three models were trained based on the previously extracted features.

Support Vector Machine (SVM): An SVM is a supervised algorithm that searches for the optimal hyperplane to separate classes by maximizing the margin between closest samples. In nonlinearly separable problems, the kernel trick is used to project data to higher-dimensional spaces, thereby achieving efficient classification even in complex scenarios [32].

$$w.x + b = 0 \tag{2}$$

Where:

- w: is the vector of weights
- x: is the vector of characteristics
- b: is the bias term
- $(.)$: represents the dot product between w and x

Random Forest (RF): The RF algorithm is a supervised learning algorithm based on the construction of multiple decision trees. Each tree was trained with different subsets of data and features randomly. The final prediction is made by majority voting of the trees (in classification) or the average of their predictions (in regression), which improves accuracy and reduces overfitting [33].

$$\hat{y}(x) = \arg \max_{y \in \{1,\dots,K\}} \sum_{j=1}^{j} I\big(h_j(x) = y\big) \tag{3}$$

Where:

- J: Represents the total number of trees in the forest
- $h_j(x)$: The prediction of the j- th tree for the input x
- $I(.)$: The indicator function, which equals 1 if the tree's prediction matches the class y, and 0 otherwise.
- K: The total number of possible classes

K- Nearest Neighbors (KNN): This supervised learning algorithm is used for both classification and regression. The proposed method is based on the notion that similar

data tend to be close to each other in feature space. To make a prediction, the KNN locates the k closest data points in the training set and makes a decision based on the majority (classification) or average (regression) of those neighbors [34].

$$d\,(p,q)\; = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2} \tag{4}$$

Where:

- p and q are two points in the feature space.
- n is the number of features.
- p_i and q_i represent the values of the features of points p and q, respectively.

Parameters for Machine Learning Classifiers.

a) Support Vector Machine (SVM):

- C: 0.1, 1, 10, 50
- kernel: linear, rbf, poly
- gamma: 0.1, 1, scale, auto

b) Random Forest (RF)

- n_estimators: 50, 10, 20
- max_depth: None, 5, 10, 20
- criterion: gini, entropy

c) K-Nearest Neighbors (KNN)

- n_neighbors: 3, 5, 7
- weights: uniform, distance
- metric: euclidean, manhattan, minkowski
- p: 1, 2

3.5 Application of Deep Learning Models

In this phase, six architectures were implemented to classify diabetic retinopathy images.

BOTNet (Bottleneck Transformer Network): It is a neural network architecture that combines convolutions with the attention mechanism of Transformers, introducing the latter in the deep layers of the model. BOTNet is based on ResNet, but instead of using only convolutions in the upper layers, it uses transformer blocks to capture long-range relationships between image pixels. In the first layers, convolutions are responsible for extracting local features, such as edges and textures, and deep layers with Transformers are responsible for capturing global features, such as the overall structure of the object in the image. The proposed method allows BOTNet to perform outstandingly well in computer vision tasks, particularly in image classification and object detection. The

combination of convolutions and attention allows BOTNet to capture both fine details and relationships at a global level, making it highly effective on complex datasets [35].

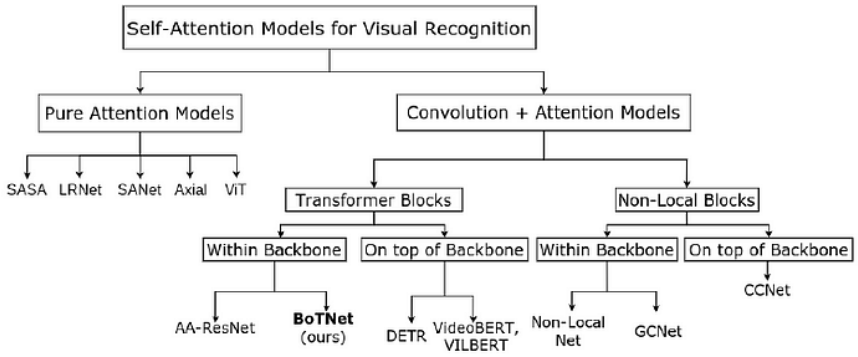


Fig. 3. The BOTNet - model architecture

CoAtNet (Convolutional-Attention Network): An enhancement to the Vision Transformer (ViT) architecture, which introduces the concept of shifted windows to address the limitations of ViT in capturing local features. Unlike ViT, which divides the entire image into large patches, Swin Transformer applies the attention of the Transformers within small local windows, allowing fine details to be captured at early stages. Then, it combines these windows to capture global features across the entire image. This design allows SwinTransformer to strike a balance between capturing local and global features, which makes it more efficient for tasks such as image classification, object detection, and segmentation. In addition, the hierarchical structure of SwinTransformer allows it to process images of different sizes more efficiently than other Transformers [36].

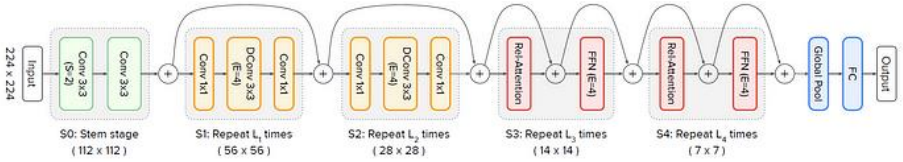


Fig. 4. The CoAtNet - model architecture

Swin Transformer: This is an enhancement of the Vision Transformer (ViT) architecture, which introduces the concept of shifted windows to address the limitations of ViT in capturing local features. Unlike ViT, which divides the entire image into large patches, Swin Transformer applies the attention of the Transformers within small local windows, allowing fine details to be captured at early stages. Then, it combines these windows to capture global features across the entire image. This design allows the Swin Transformer to strike a balance between capturing local and global features, making it more efficient for tasks such as image classification, object detection, and segmentation. In addition, the Swin Transformer's hierarchical structure allows it to process images of different sizes more efficiently than other Transformers [37].

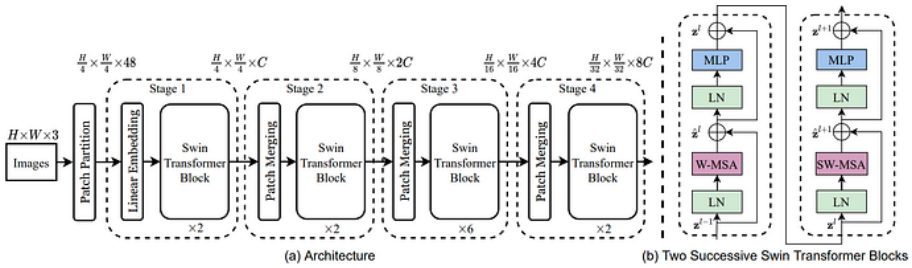


Fig. 5. The Swin Transformer – model architecture

PiT (Pooling-based Vision Transformer): It is a variant of ViT that introduces the pooling operation, common in convolutional networks, within the Transformer architecture. This model uses pooling layers to reduce the resolution of extracted features in the early stages, like what traditional convolutional networks do. This improves PiT's ability to capture local features in the early layers while maintaining global attention capability in deeper layers. As in standard Transformers, the self-attenuation mechanism in PiT allows modeling global interactions between different regions of the image, while pooling operations ensure that local details are also captured efficiently. PiT is a solid choice for tasks such as image classification, as it combines the best of CNNs and Transformers [38].

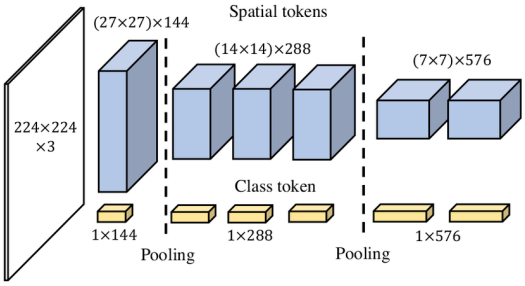


Fig. 6. The PiT – model architecture

ViT (Vision Transformer): is a model for image classification that employs a Transformer-like architecture over patches of the image. An image is split into fixed-size patches, each of them are then linearly embedded, position embeddings are added, and the resulting sequence of vectors is fed to a standard Transformer encoder. In order to perform classification, the standard approach of adding an extra learnable "classification token" to the sequence is used [39].

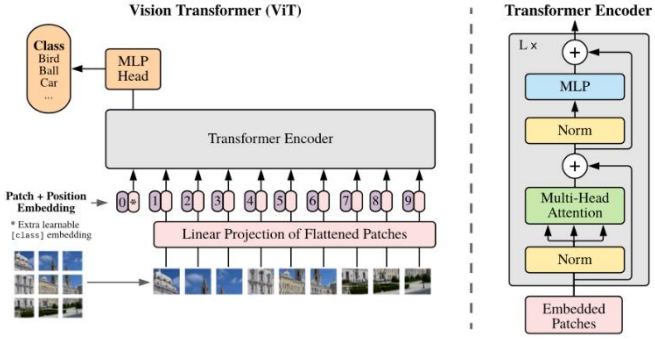


Fig. 7. The ViT – model architecture

MobileViT: A lightweight version of the vision transformer (ViT), designed to be efficient for mobile devices and resource-limited systems. MobileViT combines convolutions and Transformers in an optimized architecture to capture both local and global features. The first convolutional layers are used to extract local features and reduce the dimensionality of the image, while the Transformers layers capture the global relationships between different parts of the image. This combination allows MobileViT to perform well in computer vision tasks without sacrificing computational resources. It is ideal for applications that require memory- and runtime-efficient vision models, such as image classification and object detection on mobile devices [40].

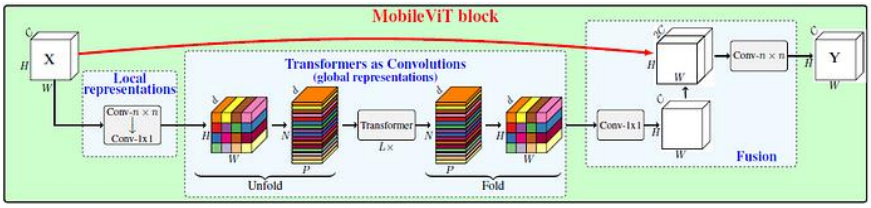


Fig. 8. The MobileViT - model architecture

Parameters for Deep Learning Classifiers

a) Bottleneck Transformers Network (BOTNet)

- learning_rate: 0.001, 0.0001
- optimizer: AdamW, Adam, SGD
- batch_size: 16, 32
- epochs: 15

b) Convolution and Attention Network (CoAtNet)

- learning_rate: 0.001, 0.0001
- optimizer: AdamW, Adam, SGD
- batch_size: 16, 32
- epochs: 15

c) Vision Transformer (ViT)

- learning_rate: 0.001, 0.0001

- optimizer: Adam, AdamW, SGD
- batch_size: 16, 32
- epochs: 15

d) Shifted Windows Transformer (SwinTransformer)

- learning_rate: 0.001, 0.0001
- optimizer: AdamW, Adam, SGD
- batch_size: 16, 32
- epochs: 15

e) Pooling-based Vision Transformer (PiT)

- learning_rate: 0.001, 0.0001
- optimizer: Adam, AdamW, SGD
- batch_size: 16, 32
- epochs: 15

f) Mobile Vision Transformer (MobileViT)

- learning_rate: 0.001, 0.0001
- optimizer: Adam, AdamW, SGD
- batch_size: 16, 32
- epochs: 15

3.6 Application of the Hybrid Approach

In this phase, the 6 deep learning models combined with the 3 machine learning classifiers were used.

Parameters for Hybrid Classifiers

a) Support Vector Machine (SVM)

- C: 0.1, 1, 10, 100
- kernel: linear. Rbf
- gamma: 0.01, 0.1, 1, scale, auto

b) Random Forest (RF)

- n_estimators: 50, 100, 200
- max_depth: None, 10, 20, 30
- criterion: gini, entropy

c) K-Nearest Neighbors (KNN)

- n_neighbors: 3, 5, 7, 9, 11, 13
- weights: uniform, distance
- metric: euclidean, manhattan, minkowsi

3.7 Model Evaluation

Once the training of the models on the corresponding dataset is completed, it is crucial to evaluate their ability to make accurate predictions on previously unobserved data. This evaluation was performed using an independent set of test images to objectively

measure model performance during the validation phase. Key metrics, such as precision, accuracy, recall, and F1-Score, were used to provide a comprehensive view of the effectiveness and predictive ability of each model.

4 Results

This section presents the best results corresponding to the proposed methodology (see Fig. 1). The set of parameters was carefully chosen to maximize the performance of each model.

4.1 Results of Machine Learning (methodology 1)

Table 3 shows that the SVM-SIFT classification model, with hyperparameters C: 1, gamma: scale and kernel: rbf, obtained the highest scores.

Table 3. Metrics of Machine Learning models

Method	Model	Accuracy	Precision	Recall	F1-Score	Hyperparameter
SIFT	SVM	**0.9264**	**0.9286**	**0.9203**	**0.9244**	C:1 gamma: scale kernel: rbf
	RF	0.8788	0.8400	0.9292	0.8889	criterion: gini max_depth: 10 n_estimators: 20
	KNN	0.8788	0.8632	0.8938	0.8783	metric: euclidean n_neighbors: 9 p: 1 weights: uniform
ORB	SVM	0.7749	0.7652	0.7788	0.7719	C: 1 gamma: scale kernel: rbf
	RF	0.7489	0.7570	0.7168	0.7558	criterion: entropy max_depth: None n_estimators: 20
	KNN	0.7273	0.7155	0.7345	0.7249	metric: euclidean n_neighbors: 7 weights: uniform
HOG	SVM	0.9221	0.9130	0.9292	0.9210	C: 0.1 gamma: 0.1 kernel: linear
	RF	0.9048	0.8699	0.9469	0.8983	criterion: entropy max_depth: None n_estimators: 20
	KNN	0.9091	0.8770	0.9469	0.9106	metric: manhattan n_neighbors: 3 p: 1 weights: distance

Based on the results obtained in Table 5, the SVM model combined with SIFT achieved the best performance across all evaluated metrics. It reached an accuracy of 92.64%, demonstrating its robustness in classifying both positive and negative cases. Its precision of 92.86% highlights its ability to minimize false positives, which is critical in medical contexts to avoid unnecessary tests or treatments. Additionally, the recall of 92.03% ensures the detection of most positive cases, essential for early diagnosis. Finally, the F1-Score of 92.44% reflects an effective balance between sensitivity and precision, confirming its suitability for reliable classification tasks.

The confusion matrix is a tool for evaluating the performance of classification models because it helps analyze errors and successes to facilitate continuous improvement [41], [42].

Fig. 9 shows the confusion matrix of the SVM-SIFT model, detailing the correct and incorrect predictions:

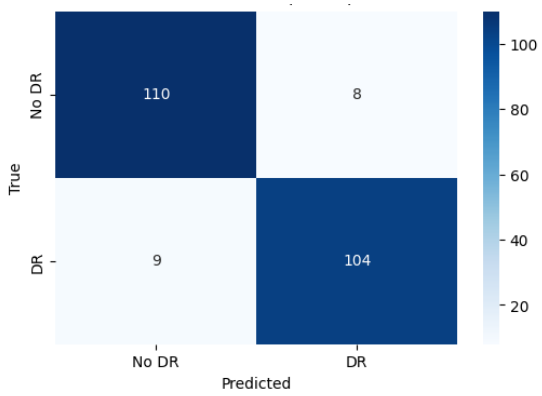


Fig. 9. Confusion Matrix: SVM (Tuned) with SIFT

- **True Negatives (TN)**: We correctly identified **110** negative cases. These are cases in which the model correctly predicted that the patient did not have Diabetic Retinopathy.
- **False Positives (FP)**: **8** cases were incorrectly identified as positive. These are cases in which the model erroneously predicted that the patient had diabetic retinopathy when, in fact, he did not.
- **False Negatives (FN)**: **9** cases were incorrectly identified as negative. These are cases in which the model erroneously predicted that the patients did not have diabetic retinopathy when, in fact, they did.
- **True Positives (TP)**: We correctly identified **104** positive cases. These are cases in which the model correctly predicted that the patient had Diabetic Retinopathy.

4.2 Results of Deep Learning (methodology 2)

Table 4 shows PiT as the best performing model after applying the hyperparameters: batch_size: 32, epochs: 15, learning_rate: 0.001 and optimizer: SGD.

Table 4. Metrics of Deep Learning models

Model	Accuracy	Precision	Recall	F1-Score	Hyperparameter
BOTNet	0.9784	0.9821	0.9735	0.9778	batch_size: 32 epochs: 15 learning_rate: 0.0001 optimizer: Adam
CoAtNet	0.9784	0.9821	0.9735	0.9778	batch_size: 16 epochs: 15 learning_rate: 0.0001 optimizer: SGD
ViT	0.9784	0.9821	0.9735	0.9778	batch_size: 16 epochs: 15 learning_rate: 0.0001 optimizer: SGD
Swin-Trans-former	0.9740	0.9820	0.9646	0.9732	batch_size: 16 epochs: 15 learning_rate: 0.0001 optimizer: SGD
PiT	**0.9870**	**0.9911**	**0.9823**	**0.9867**	batch_size: 32 epochs: 15 learning_rate: 0.001 optimizer: SGD
MobileVit	0.9784	0.9821	0.9735	0.9778	batch_size: 32 epochs: 15 learning_rate: 0.0001 optimizer: AdamW

The results from Table 7 show that the PiT model achieved the best performance among all evaluated models, with the highest Accuracy (98.70%), Precision (99.11%), Recall (98.23%), and F1-Score (98.67%). These metrics confirm its robustness and reliability for classification tasks, excelling at minimizing false positives and detecting the majority of positive cases. The configuration of the model, which includes a batch size of 32, a learning rate of 0.001, and the SGD optimizer, played a key role in achieving its superior performance.

Fig. 10 presents the results, showing the correct and incorrect predictions of the PiT model with the highest metric values.

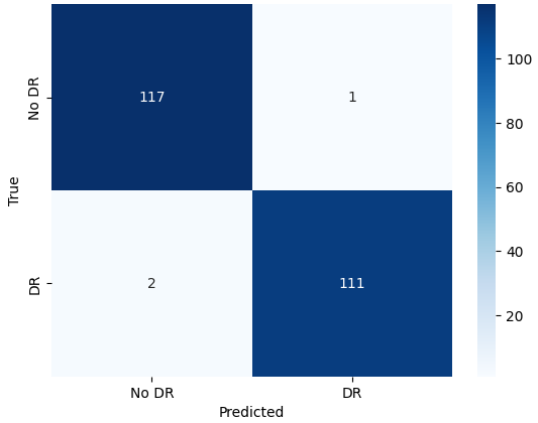


Fig. 10. Confusion Matrix: PiT (tuned).

Fig. 11 shows the loss and accuracy curves during the training of the PiT model.

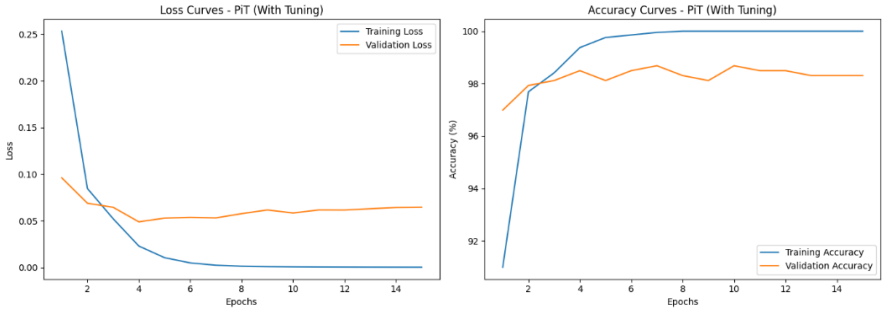


Fig. 11. Accuracy and Loss curves.

The loss curves showed a continuous decrease in both the training and validation losses, with the validation loss stabilizing at low values after a few epochs, which indicates that the model generalized after hyperparameter tuning. For the accuracy curves, training accuracy quickly reached 100%, and validation accuracy stabilized at approximately 98%, with minimal fluctuations.

4.3 Results of Hybrid (methodology 3)

This part of the methodology was based on a hybrid approach (Deep Learning and Machine Learning). The best results were obtained with RF-PiT, without resorting to hyperparameters.

Table 5. Metrics of Hybrid models

Feature Extractor	Model	Accuracy	Precision	Recall	F1-Score
BOTNet	SVM	0.9654	0.9730	0.9557	0.9643
	RF	0.9567	0.9557	0.9557	0.9557
	KNN	0.9654	0.9646	0.9646	0.9646
CoAtNet	SVM	0.9697	0.9818	0.9557	0.9686
	RF	0.9654	0.9646	0.9646	0.9646
	KNN	0.9610	0.9815	0.9380	0.9593
ViT	SVM	0.9697	0.9732	0.9646	0.9689
	RF	0.9567	0.9640	0.9469	0.9553
	KNN	0.9567	0.9640	0.9469	0.9553
SwinTransformer	SVM	0.9437	0.9464	0.9380	0.9422
	RF	0.9740	0.9734	0.9734	0.9734
	KNN	0.9697	0.9732	0.9646	0.9689
PiT	SVM	0.9437	0.9386	0.9469	0.9427
	RF	**0.9827**	**0.9910**	**0.9734**	**0.9821**
	KNN	0.9697	0.9818	0.9557	0.9686
MobileViT	SVM	0.9394	0.9380	0.9380	0.9380
	RF	0.9610	0.9643	0.9557	0.9600
	KNN	0.9480	0.9549	0.9380	0.9464

Table 5 presents the performance of different hybrid models using various feature extractors. Among them, the RF model combined with PiT stands out as the best, achieving the highest accuracy (98.27%), precision (99.10%), recall (97.34%) and F1-Score (98.21%). These results demonstrate that a hybrid approach combining Deep Learning and Machine Learning can achieve exceptional performance without relying on the tedious search for the optimal set of hyperparameters.

Fig. 12 presents the results, displaying the confusion matrix of the model with the highest metric values, showing both correct and incorrect predictions.

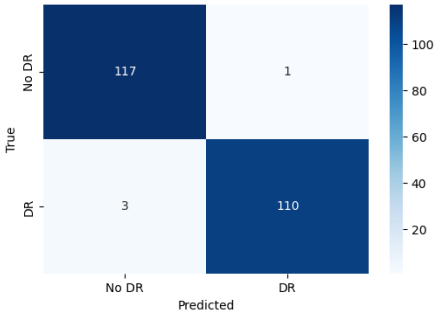


Fig. 12. Confusion matrix: RF with PiT

5 Conclusions

In this study, diabetic retinopathy was identified as a significant threat to the vision of patients with diabetes mellitus, affecting millions of people worldwide. The importance of early detection was highlighted as a crucial measure to prevent disease progression. The role of robust deep learning-based models was emphasized as a highly effective solution to overcome traditional visual inspection methods, significantly improving accuracy and efficiency in the classification and early diagnosis of this condition.

Data acquisition was performed using a set downloaded from the Kaggle website containing color fundus images with 224 × 224-pixel resolution, organized in hierarchical folders for training, validation, and testing.

During preprocessing, advanced techniques designed to optimize image quality were applied, thereby reducing the risk of overfitting and improving the learning capacity and accuracy of the models. In Methodology 1, techniques such as CLAHE and grayscale conversion were implemented to highlight details and improve image contrast, which facilitated feature extraction using SIFT, ORB, and HOG. In Methodology 2, standardization was applied to normalize the pixel intensity, thereby achieving a uniform distribution of the data and optimizing the training of the models for classification tasks.

For classification, three methodological approaches were explored: auto-mathematical learning (SVM, RF and KNN), deep learning (BOTNet, CoAtNet, SWinTrans-former, PiT, ViT and MobileViT), and hybrid approaches. Among these, the deep learning-based methodology proved to be the most efficient, especially the PiT model, which achieved an outstanding performance with an accuracy of 98.70%, an accuracy of 99.11%, a recall of 98.23%, and an F1-Score of 98.27. These results highlight the ability of deep learning models to extract complex features automatically and efficiently, making them highly robust tools for clinical applications. Although the machine learning and hybrid approaches provided competitive results, the overall analysis confirmed that the deep learning models, which were optimized using hyperparameters, represented the most robust solution. In particular, the hyperparameters applied to the PiT model included a batch size of 32, 15 epochs, a learning rate of 0.001, and the SGD optimizer.

In conclusion, this study demonstrated the potential of deep learning, particularly the hyperparameter-optimized PiT model, for early detection of diabetic retinopathy. The results highlight the applicability of the proposed method to automated clinical systems capable of rapid and reliable diagnosis, even in regions with limited resources. Furthermore, this study provides a foundation for future research to integrate advanced technologies and improve early diagnosis and personalized treatment.

References

1. M. Kropp *et al.*, “Diabetic retinopathy as the leading cause of blindness and early predictor of cascading complications—risks and mitigation,” 2023. doi: 10.1007/s13167-023-00314-8.
2. American Optometric Association, “Diabetic retinopathy,” American Optometric Association (AOA). Accessed: Sep. 27, 2024. [Online]. Available: https://www.aoa.org/healthy-eyes/eye-and-vision-conditions/diabetic-retinopathy?sso=y

3. Merck Manuals, “Diabetic Retinopathy,” Merck Manual: Professional Version. Accessed: Sep. 27, 2024. [Online]. Available: https://www.merckmanuals.com/professional/eye-disorders/retinal-disorders/diabetic-retinopathy?query=diabetic%20retinopathy
4. M. Mrugacz, A. Bryl, and K. Zorena, “Retinal Vascular Endothelial Cell Dysfunction and Neuroretinal Degeneration in Diabetic Patients,” *J Clin Med*, vol. 10, no. 3, p. 458, Jan. 2021, doi: 10.3390/jcm10030458.
5. Z. L. Teo *et al.*, “Global Prevalence of Diabetic Retinopathy and Projection of Burden through 2045,” *Ophthalmology*, vol. 128, no. 11, pp. 1580–1591, Nov. 2021, doi: 10.1016/j.ophtha.2021.04.027.
6. Visian Atlas, “Diabetic Retinopathy,” International Agency for the Prevention of Blindness (IAPB). Accessed: Sep. 27, 2024. [Online]. Available: https://www.iapb.org/learn/vision-atlas/causes-of-vision-loss/diabetic-retinopathy/
7. World Health Organization, “Blindness and vision impairment,” World Health Organization (WHO). Accessed: Nov. 03, 2024. [Online]. Available: https://www.who.int/es/news-room/fact-sheets/detail/blindness-and-visual-impairment
8. M. L. Avilés-Santa, A. Monroig-Rivera, A. Soto-Soto, and N. M. Lindberg, “Current State of Diabetes Mellitus Prevalence, Awareness, Treatment, and Control in Latin America: Challenges and Innovative Solutions to Improve Health Outcomes Across the Continent,” *Curr Diab Rep*, vol. 20, no. 11, p. 62, Nov. 2020, doi: 10.1007/s11892-020-01341-9.
9. J. E. Villena, “Diabetes Mellitus in Peru,” *Ann Glob Health*, vol. 81, no. 6, pp. 765–775, 2015, doi: 10.1016/j.aogh.2015.12.018.
10. Instituto Nacional de Estadistica e Informatica, “El 4,5% de las personas de 15 y más años de edad tiene la diabetes mellitus,” Instituto Nacional de Estadistica e Informatica (INEI). Accessed: Sep. 27, 2024. [Online]. Available: https://m.inei.gob.pe/prensa/noticias/el-399-de-peruanos-de-15-y-mas-anos-de-edad-tiene-al-menos-una-comorbilidad-12903/
11. Kierstan Boyd, “Diabetic Retinopathy Diagnosis,” American Academy of Ophthalmology (AAO).
12. J. G. Fujimoto, E. A. Swanson, and D. Huang, “Optical Coherence Tomography—History, Evolution, and Future Prospects,” *JAMA*, vol. 330, no. 15, pp. 1427–1428, 2023, doi: 10.1001/jama.2023.16942.
13. X. Feng, Y. Cai, and R. Xin, “Optimizing diabetes classification with a machine learning-based framework,” *BMC Bioinformatics*, vol. 24, no. 1, 2023, doi: 10.1186/s12859-023-05467-x.
14. T. Sharma and M. Shah, “A comprehensive review of machine learning techniques on diabetes detection,” *Vis Comput Ind Biomed Art*, vol. 4, no. 1, 2021, doi: 10.1186/s42492-021-00097-7.
15. D. Das, S. K. Biswas, and S. Bandyopadhyay, “Detection of Diabetic Retinopathy using Convolutional Neural Networks for Feature Extraction and Classification (DRFEC),” *Multimed Tools Appl*, vol. 82, no. 19, pp. 29943–30001, 2023, doi: 10.1007/s11042-022-14165-4.

16. H. S. Alghamdi, “Towards Explainable Deep Neural Networks for the Automatic Detection of Diabetic Retinopathy,” *Applied Sciences (Switzerland)*, vol. 12, no. 19, 2022, doi: 10.3390/app12199435.
17. C. Mohanty *et al.*, “Using Deep Learning Architectures for Detection and Classification of Diabetic Retinopathy,” *Sensors*, vol. 23, no. 12, 2023, doi: 10.3390/s23125726.
18. A. Ayala, T. Ortiz Figueroa, B. Fernandes, and F. Cruz, “Diabetic retinopathy improved detection using deep learning,” *Applied Sciences (Switzerland)*, vol. 11, no. 24, 2021, doi: 10.3390/app112411970.
19. M. M. Butt, D. N. F. A. Iskandar, S. E. Abdelhamid, G. Latif, and R. Alghazo, “Diabetic Retinopathy Detection from Fundus Images of the Eye Using Hybrid Deep Learning Features,” *Diagnostics*, vol. 12, no. 7, 2022, doi: 10.3390/diagnostics12071607.
20. D. Hassan, H. M. Gill, M. Happe, A. D. Bhatwadekar, A. R. Hajrasouliha, and S. C. Janga, “Combining transfer learning with retinal lesion features for accurate detection of diabetic retinopathy,” *Front Med (Lausanne)*, vol. 9, Nov. 2022, doi: 10.3389/fmed.2022.1050436.
21. A. S. Sathwik, R. Agarwal, A. Jubilson E, and S. S. Basa, “Diabetic Retinopathy Classification Using Deep Learning,” *EAI Endorsed Trans Pervasive Health Technol*, vol. 9, Nov. 2023, doi: 10.4108/eetpht.9.4335.
22. S. Guefrachi, A. Echtioui, and H. Hamam, “Diabetic Retinopathy Detection Using Deep Learning Multistage Training Method,” *Arab J Sci Eng*, May 2024, doi: 10.1007/s13369-024-09137-9.
23. B. B. Sinha, R. Dhanalakshmi, and K. Balakrishnan, “Early diagnosis of diabetic retinopathy using deep learning techniques,” in *Computational Methods and Deep Learning for Ophthalmology*, Elsevier, 2023, pp. 17–33. doi: 10.1016/B978-0-323-95415-0.00006-1.
24. D. Muthusamy and P. Palani, “Deep learning model using classification for diabetic retinopathy detection: an overview,” *Artif Intell Rev*, vol. 57, no. 7, p. 185, Jun. 2024, doi: 10.1007/s10462-024-10806-2.
25. S. S N, A. M, and S. S, “Deep Learning-Based Diabetic Retinopathy Severity Classification and Progression Time Estimation,” *IFAC-PapersOnLine*, vol. 58, no. 3, pp. 78–83, 2024, doi: 10.1016/j.ifacol.2024.07.129.
26. K. Zuiderveld, “Contrast Limited Adaptive Histogram Equalization,” in *Graphics Gems*, Elsevier, 1994, pp. 474–485. doi: 10.1016/B978-0-12-336156-1.50061-6.
27. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*. 2016. Accessed: Oct. 19, 2024. [Online]. Available: https://books.google.com.pe/books/about/Deep_Learning.html?id=Np9SDQAAQBAJ&redir_esc=y
28. Francois Chollet, *Deep Learning with Python*, Second Edition. 2021. Accessed: Nov. 28, 2024. [Online]. Available: https://books.google.com.pe/books/about/Deep_Learning_with_Python_Second_Edition.html?id=XHpKEAAAQBAJ&redir_esc=y
29. D. G. Lowe, “Distinctive Image Features from Scale-Invariant Keypoints,” *Int J Comput Vis*, vol. 60, no. 2, pp. 91–110, Nov. 2004, doi: 10.1023/B:VISI.0000029664.99615.94.
30. E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *2011 International Conference on Computer Vision*, IEEE, Nov. 2011, pp. 2564–2571. doi: 10.1109/ICCV.2011.6126544.

31. N. Dalal and B. Triggs, “Histograms of Oriented Gradients for Human Detection,” in *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05)*, IEEE, pp. 886–893. doi: 10.1109/CVPR.2005.177.
32. Christopher M. Bishop, *Pattern Recognition and Machine Learning*. 2006. Accessed: Oct. 20, 2024. [Online]. Available: https://link.springer.com/book/9780387310732#about-authors
33. Y. Liu, Y. Wang, and J. Zhang, “New Machine Learning Algorithm: Random Forest,” 2012, pp. 246–252. doi: 10.1007/978-3-642-34062-8_32.
34. P. Cunningham and S. J. Delany, “k-Nearest Neighbour Classifiers - A Tutorial,” *ACM Comput Surv*, vol. 54, no. 6, pp. 1–25, Jul. 2022, doi: 10.1145/3459665.
35. A. Srinivas, T.-Y. Lin, N. Parmar, J. Shlens, P. Abbeel, and A. Vaswani, “Bottleneck Transformers for Visual Recognition,” arXiv preprint arXiv:2101.11605, 2021.
36. Z. Dai, H. Liu, Q. V Le, and M. Tan, “CoAtNet: Marrying Convolution and Attention for All Data Sizes,” arXiv preprint arXiv:2106.04803, 2021.
37. Z. Liu *et al.*, “Swin Transformer: Hierarchical Vision Transformer using Shifted Windows,” arXiv preprint arXiv:2103.14030, 2021.
38. H. Ren, J. Guo, S. Cheng, and Y. Li, “Pooling-based Visual Transformer with low complexity attention hashing for image retrieval,” *Expert Syst Appl*, vol. 241, p. 122745, May 2024, doi: 10.1016/j.eswa.2023.122745.
39. Alexey Dosovitskiy *et al.*, “Vision Transformer,” Papers With Code. Accessed: Jan. 29, 2025. [Online]. Available: https://paperswithcode.com/method/vision-transformer
40. Sachin Mehta and Mohammad Rastegari, “MobileViT: Light-weight, General-purpose, and Mobile-friendly Vision Transformer,” arXiv preprint arXiv:2110.02178, 2022.
41. Andreas C. Müller and Sarah Guido, *Introduction to Machine Learning with Python*. O’Reilly Media, 2016. Accessed: Oct. 25, 2024. [Online]. Available: https://www.google.com.pe/books/edition/Introduction_to_Machine_Learning_with_Py/vbQlDQAAQBAJ?hl=en&gbpv=0
42. Kyle Gallatin and Chris Albon, “Machine Learning with Python Cookbook,” Google Libros. Accessed: Oct. 21, 2024. [Online]. Available: https://www.google.com.pe/books/edition/Machine_Learn-ing_with_Python_Cookbook/WK3NEAAAQBAJ?hl=es-419&gbpv=0